CS341 · COMPUTER GRAPHICS

What Repeats 472 · 461 · 451

Questions repeated across Dr. Adil's midterms

Muharib

What's inside

01 The midterm format

02 Came in all three

03 Came twice

04 Asked once, worth memorizing

05 The problems

01 The midterm format

Part	472	461	451
Definitions and short answers	✓	✓	✓
Lists (write four)		✓	✓
Comparison	✓	✓	✓
True / False	✓	✓	
MCQ	✓	✓	✓
Aspect ratio problem		✓	
Line-drawing problem	DDA	Bresenham	Bresenham

This term the line problem is DDA; Bresenham is not included. Midterm 462 is from another doctor with a different syllabus, so it is left out of this analysis.

02 Came in all three

Question	Where	Answer
Define computer graphics	472 · 461 · 451	**Creation, storage, and manipulation of models and images**
What does a CRT emit?	472 · 461 · 451	**Electrons**
CG applications	472 · 461 · 451	**All of the above**

03 Came twice

Question	Where	Answer
Where is the raster picture stored?	472 · 451	Frame buffer or refresh buffer
Refresh rate	461 · 451	How many times per second the picture is redrawn
Write four hard-copy devices	461 · 451	Laser · Ink-jet · Film recorder · Electrostatic · Pen plotter
Absolute vs relative	472 · 461	Absolute = the actual position from the origin. Relative = an offset from the last position.
Aspect ratio	472 · 461	Keeps shapes intact: a square stays a square, a circle does not become an ellipse. $AR = \frac{\frac{Yp}{Ymax}}{\frac{Xp}{Xmax}}$
Raster vs Random	472 · 461	Random draws lines directly with the beam. Raster scans top to bottom, one horizontal line at a time.
"Applications are only for entertainment"	461 · 462	**False**
Sum of two points and Point – Point	472 · 451	Point + Point is not a valid point; Point – Point = Vector

04 Asked once, worth memorizing

Question	Where	Answer
Emissive vs nonemissive	451	Emissive converts electricity into light (plasma, LED). Nonemissive uses outside light (LCD).
Write four video display devices	451	CRT · LCD · Plasma · Thin-film electroluminescent
Active-matrix	451	Each pixel is controlled by its own transistor
A screen point is called	451	**Pixel**
Raster images are called	451	**Bitmap**
Image quality depends on	451	The number of pixels; more is better
Screen clarity depends on	472	Dot pitch
Which is an input device?	472	Mouse
NOT a main task	472	Data encryption
Goal of CG	461	Synthetic images indistinguishable from reality
Resolution	461	The maximum number of points displayed on a CRT without overlap

Persistence	461	The time for the emitted light to decay to one-tenth of its original intensity
Define a vector; from P to Q	472	Magnitude and direction. `v = Q − P`

True / false

Statement	Answer
A vector has both magnitude and direction.	**True**
Sum two points results in a valid point.	**False**
The dot product of two perpendicular vectors is equal to zero.	**True**
A unit vector has a length greater than 1.	**False**
Raster scan scans the screen from vertical to horizontal.	**False**
Raster scan is more efficient than random scan in drawing lines.	**False**
Plasma displays are more energy-efficient than LCD displays.	**False**

05 The problems

1. DDA (5 marks): the calculations, the full table and the drawing. Watch slopes above 1: y increases by 1 and x by $\frac{1}{m}$.
2. Aspect ratio (1 mark): vertical is always on top.